Exercise 4: Optimal power flow

1. **Optimal power flow**

Consider the three-bus power system depicted in Figure 1, which comprises one wind farm (WP) and three conventional generators (G_1, G_2, G_3). Wind power production cost is zero, while the marginal production cost C_i of conventional units is given in Table 1, along with the lower and upper generation capacity limits $\underline{p}_i$ and $\overline{p}_i$ (in MW), respectively. The transmission capacities and the reactances of all transmission lines are equal to 150 MW and 0.13 p.u. (with $S_{\text{base}} = 100$ MVA), respectively. In this problem we consider hourly time periods and we set node n_3 as the system reference bus.

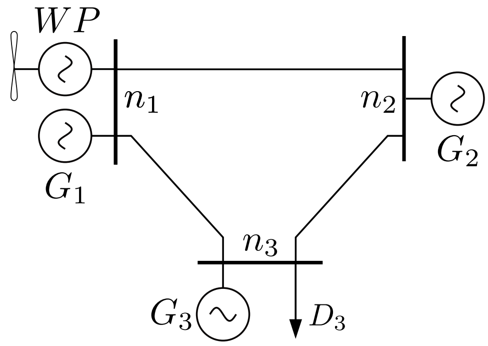

Figure 1: Three-bus power system.

Unit i	C_i (€/MWh)	$\underline{p}_i$ (MW)	$\overline{p}_i$ (MW)	$\overline{q}_i$ (MVar)
G_1	18	0	100	100
G_2	20	0	50	50
G_3	24	0	50	50

Table 1: Generator data.

Period t	1	2	3	4	5
WP	50	75	85	55	30

Table 2: Expected wind power trajectory for periods $t = 1, ..., 5$. Values in MW.

(a) Write the mathematical formulation for a single-period OPF and solve the problem to obtain the unit dispatch and locational marginal prices (LMPs), for electricity demand $D_3 = 190$ MW and wind power production $WP = 50$ MW. Check Matlab code `Exercise4_DC_StudentVersion.m` in Moodle and fill in the missing parts indicated by '`% !!!`'. **Tip!** Consider the possibility of wind power curtailment.

(b) Reduce the transmission capacity limit of line (n_1, n_3) to 100 MW and solve again the single-period OPF. Is there any change to unit dispatch and LMPs? If so, how can you explain this change?

(c) The trajectory of expected wind power production for the next five periods ($t = 1, ..., 5$) is provided in Table 2. Return transmission capacity limits of line (n_1, n_3) to 150 MW and use the expected wind power trajectory to solve a multi-period OPF for $D_3 = 200$ MW throughout the scheduling horizon. Report generation dispatch and LMPs, and comment on those results.

(d) Consider that unit G_1 is an inflexible unit that cannot change its set-point within the five-period optimization horizon. Units G_2 and G_3 are flexible generators with symmetric upward and downward ramping limits of ± 10 MW per period. Assume that the initial power output (P_i^{ini}) at time $t = 0$ of the generators is equal to $P_{i_1}^{\text{ini}} = 100$ MW, $P_{i_2}^{\text{ini}} = 45$ MW and $P_{i_3}^{\text{ini}} = 20$ MW. Include these ramping constraints in your multi-period OPF and obtain the new dispatch of generating units. Comment on the effect of ramping constraints on dispatch and LMPs.

(e) Repeat question (a) using the AC power flow equations. Assume that all transmission line resistances is 0.013 p.u. and the reactive power demand is 40 MVar. The reactive power limits of the conventional generators are $-\overline{q}_i \leq q_i \leq \overline{q}_i$ while the voltage limits are $0.95 \leq |v_i| \leq 1.05$. The wind farm also injects reactive power with a power factor of 0.95 and the system reference voltage at n_3 is 1 p.u. Comment on the results compared to question (a). Check Matlab code `Exercise4_AC_StudentVersion.m` in Moodle and fill in the missing parts indicated by '`% !!!`'.